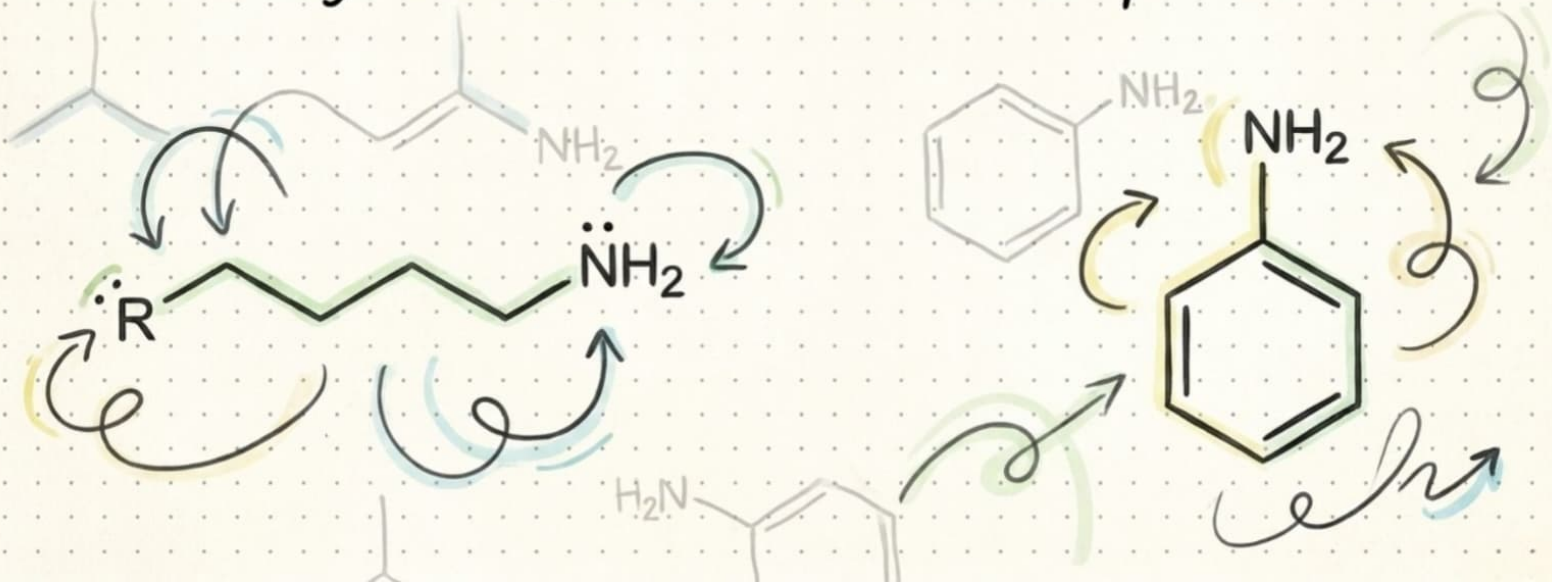
Amines: Complete Study Notes

Synthesis, Reactions, and Properties



$R-NO_2 \xrightarrow{H_2/Pd,\ ethanol\ \text{OR}\ Sn+HCl}$

$NH_3 + R-X \longrightarrow$

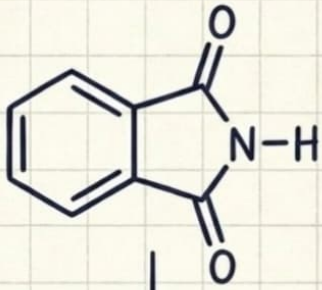
$R-C\equiv N \xrightarrow{H_2/Ni,\ Na(Hg)/C_2H_5OH}$

$R-\overset{\overset{\displaystyle O}{\|}}{C}-NH_2 \xrightarrow{(i)\ LiAlH_4\ \ (ii)\ H_2O}$

$R-NH_2$

Key Name Reactions!

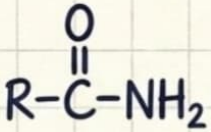
Gabriel Phthalimide Synthesis



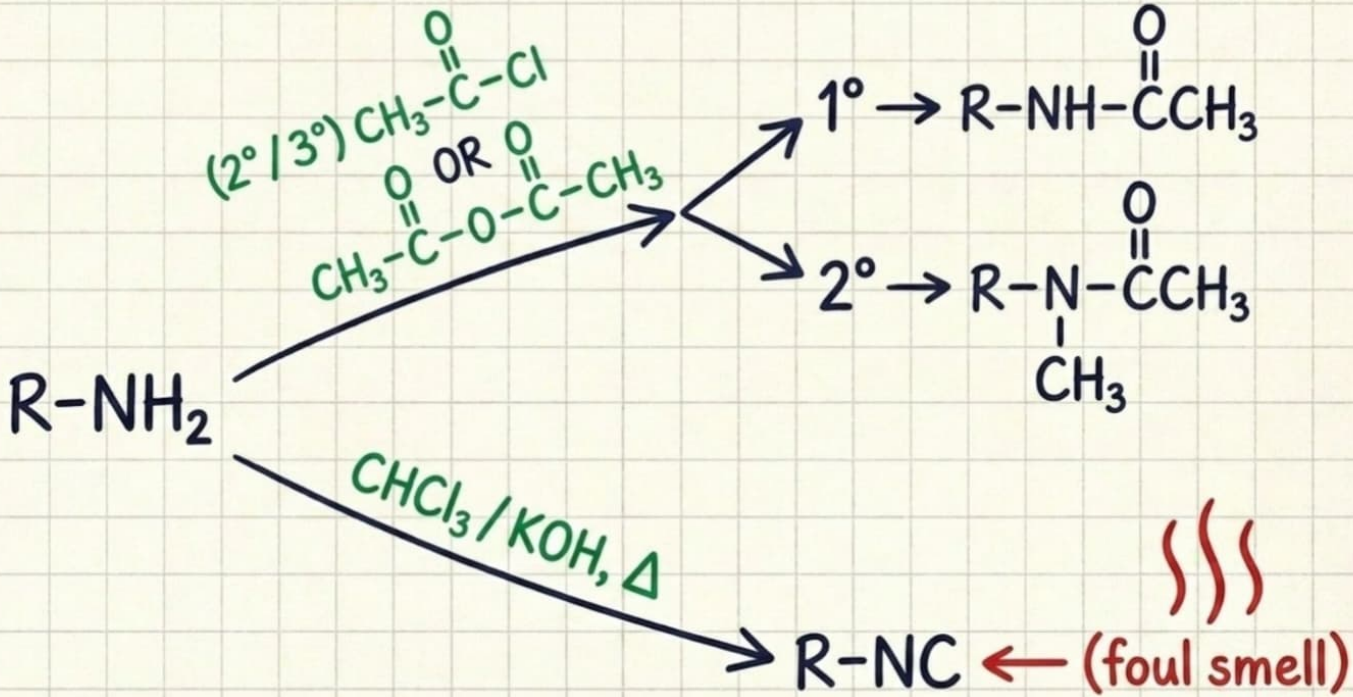
(i) KOH
(ii) R–X
(iii) NaOH

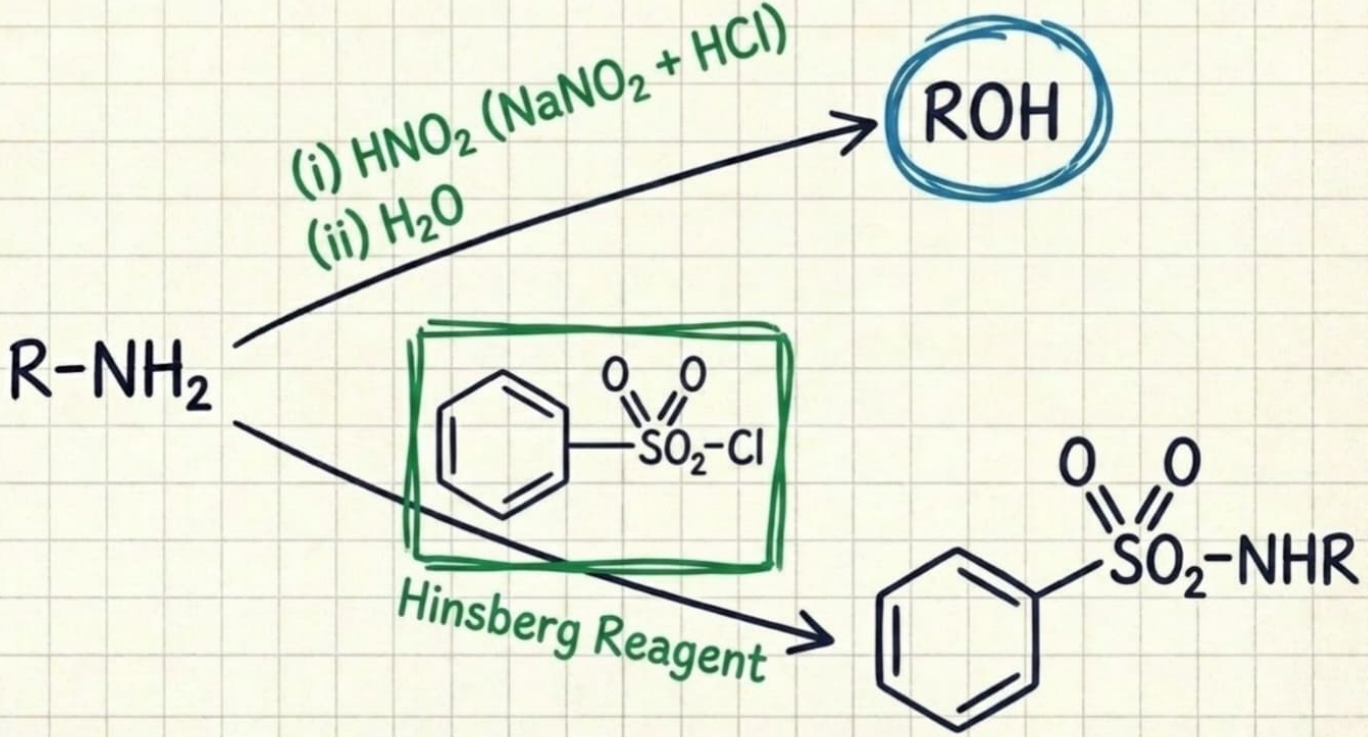
$R\text{–}NH_2$

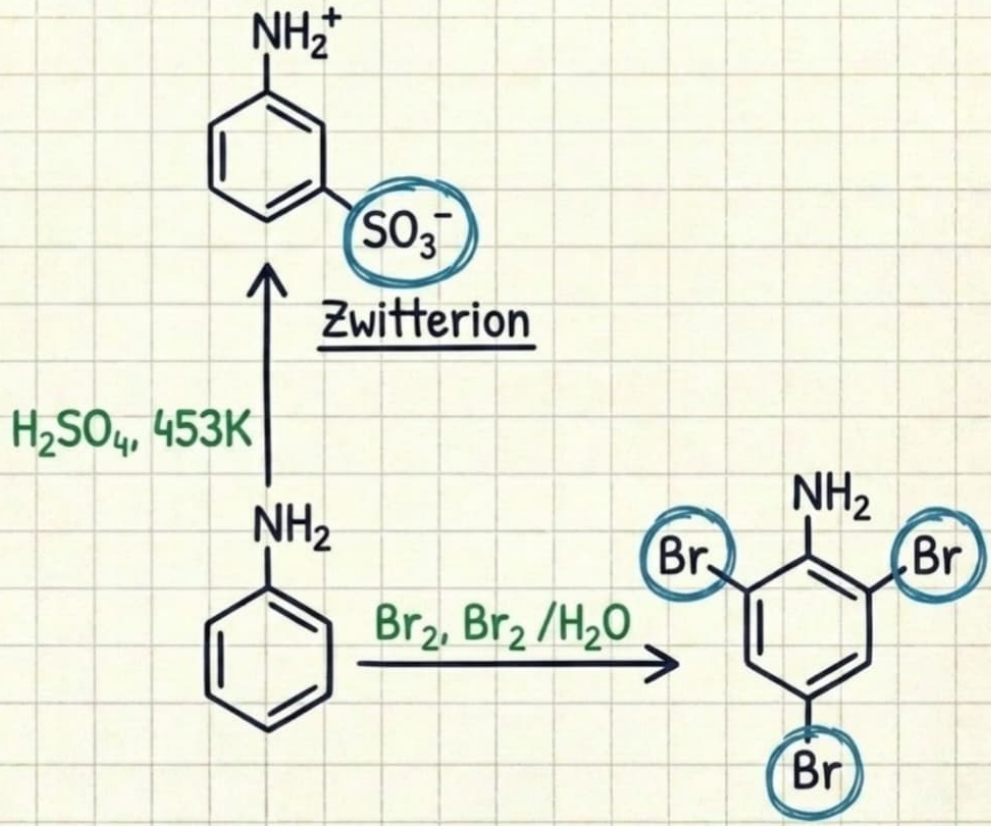
Hoffmann Bromamide Degradation

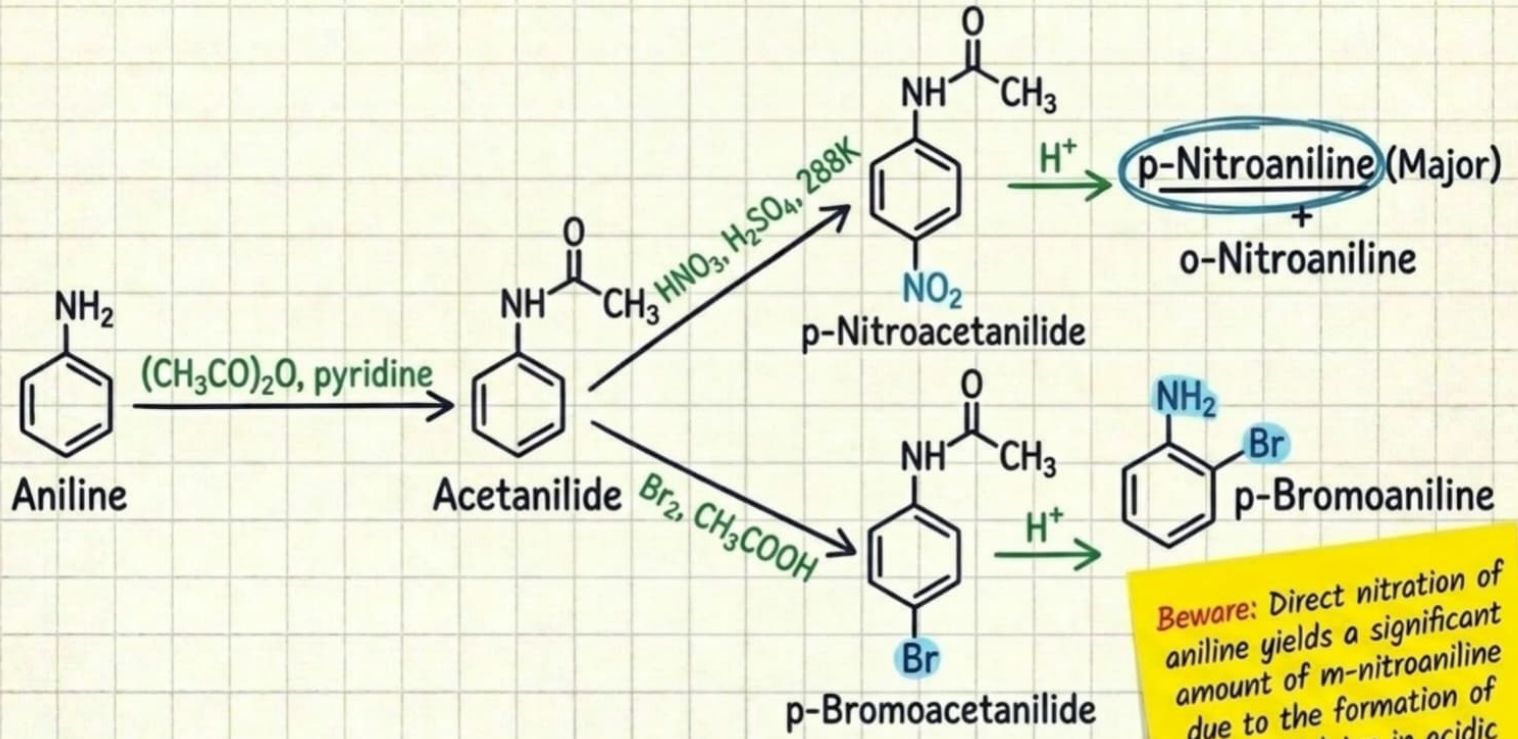


$\xrightarrow{Br_2 + NaOH(aq)}$ $\boxed{R\text{–}NH_2}$

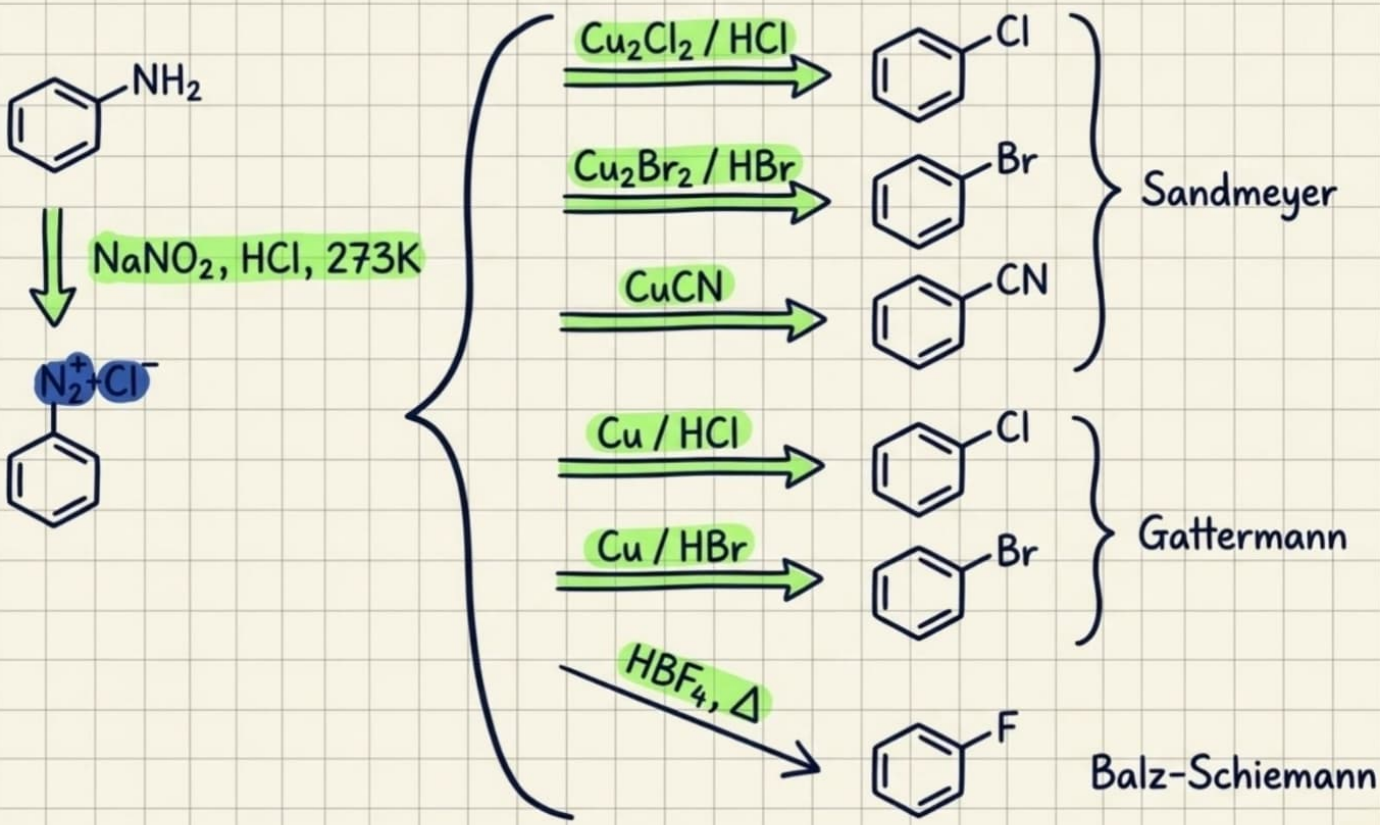

(2° / 3°) CH3-C(=O)-Cl
OR
CH3-C(=O)-O-C(=O)-CH3
R-NH2
1° → R-NH-C(=O)CH3
2° → R-N(CH3)-C(=O)CH3
CHCl3 / KOH, Δ
R-NC
(foul smell)

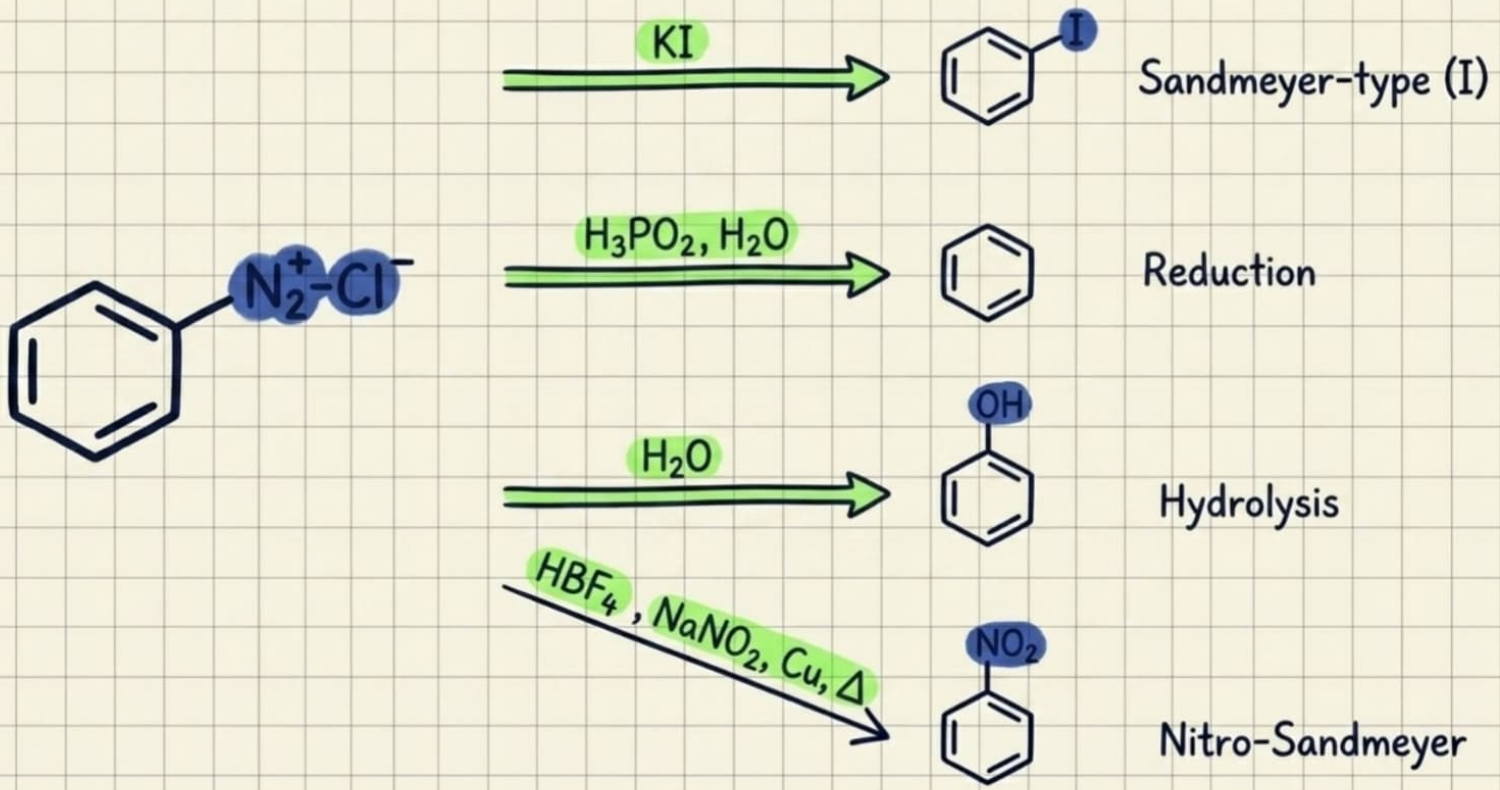

(i) HNO_2 ($NaNO_2$ + HCl)
(ii) H_2O
ROH
R-NH_2
O O
SO_2-Cl
Hinsberg Reagent
O O
SO_2-NHR

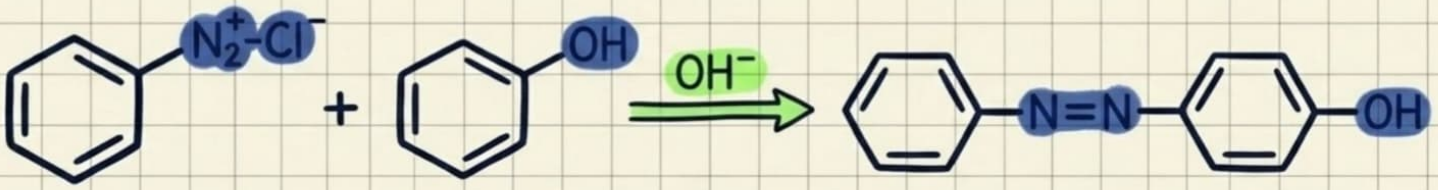

NH2+
SO3−
Zwitterion
H2SO4, 453K
NH2
Br2, Br2 /H2O
NH2
Br
Br
Br



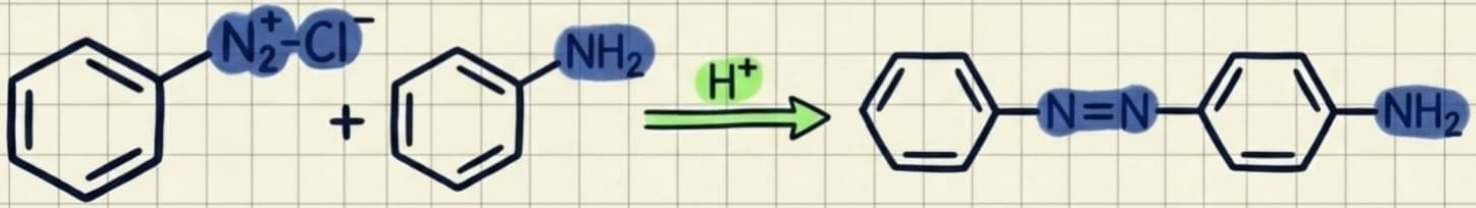
Beware: Direct nitration of aniline yields a significant amount of m-nitroaniline due to the formation of the $-NH_3^+$ ion in acidic medium!


NH2
NaNO2, HCl, 273K
N2+Cl−
Cu2Cl2 / HCl
Cl
Cu2Br2 / HBr
Br
CuCN
CN
Sandmeyer
Cu / HCl
Cl
Cu / HBr
Br
Gattermann
HBF4, Δ
F
Balz-Schiemann


N_2^+-Cl^-
KI
I
Sandmeyer-type (I)
H_3PO_2, H_2O
Reduction
H_2O
OH
Hydrolysis
HBF_4, $NaNO_2$, Cu, Δ
NO_2
Nitro-Sandmeyer



p-Hydroxyazobenzene (Yellow dye)



p-Aminoazobenzene (Orange dye)

Why are they basic?

- Basic character increases due to the presence of lone pairs (lp), methyl groups, +I effect, and Electron Donating Groups (EDG).

Basicity Trends (Aqueous Phase)

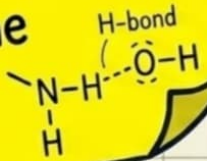
For Methyl groups: $2^\circ > 1^\circ > 3^\circ$
For other groups (Ethyl): $2^\circ > 3^\circ > 1^\circ$

Basicity Trends (Gaseous Phase)

For all groups: $3^\circ > 2^\circ > 1^\circ$

- Arylamines:
  In aniline or arylamines, the ion formed is stabilised.

1° & 2° amines can form H-bonds. 3° amines cannot due to the lack of an H-atom on the Nitrogen.

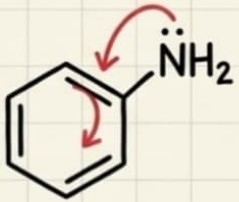


Amines are soluble in water. However, solubility decreases with an increase in the size of the bulky alkyl groups.

Boiling Point Order:

1° > 2° > 3°.

Basicity Showdown: Amines vs. Ammonia

Alkanamines vs. Ammonia	Arylamines vs. Ammonia
• Alkanamines possess a +I effect (alkyl groups are Electron Donating Groups). • This increases electron density on the Nitrogen. • Ammonia does not have R groups, so there is no EDG effect. Therefore, Alkanamines are more basic than ammonia.	• Arylamines are weaker bases. • The lone pair on the $-NH_2$ group attached directly to the benzene ring gets involved in resonance. • Thus, the lone pair remains less available for donation. 

The Ultimate Summary Matrix

Molecule Class	Ultimate Biological Purpose
Carbohydrates →	The Energy Fuel & Structural Walls.
Proteins →	The Physical Machinery & Biocatalysts (Enzymes).
Nucleic Acids →	The Instruction Manuals (DNA/RNA).
Vitamins →	The Maintenance Crew.
Hormones →	The Communication Network.

"It is the harmonious and synchronous progress of chemical reactions in the body which leads to life."